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User Hand Book

For

Ruggedized Field Telephone RFT1001

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(DS Cat No.: _____)

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RECORD OF CHANGES

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ASSOCIATED DOCUMENTS

Technical Literature for IFT1001 comprises of the following:

- **USER HANDBOOK**
- TECHNICAL MANUAL
 - PART I: TECHNICAL INFORMATION
 - Volume 1: Technical Description
 - Volume 2: Drawings
 - PART II: MAINTENANCE
 - PART III: OVERHAUL AND RECONDITIONING INSTRUCTIONS
 - PART IV: MANUFACTURER'S PARTS LIST
 - Volume 1: Parts List
 - Volume 2: Illustrations

SAFETY WARNING

The voltages employed in this Equipment are sufficiently high to endanger human life.

Power must be switched off before servicing the equipment & great care taken when making internal adjustments etc.

Ringer output Voltage in this equipment is sufficiently high to endanger human life. Power of external battery connections / internal batteries is to be removed before servicing the equipment.

Do not touch the metal parts of the line while the System Switch is in Magneto position and Magneto Call button is pressed.

FOR FIRST AID IN CASE OF ELECTRIC SHOCK SEE OVERLEAF



FIG. 2.



FIG. 3.



FIG. 4.



1. SWITCH OFF. If this is not possible, PROTECT YOURSELF with dry insulating material and pull the victim clear of the conductor. DO NOT TOUCH THE VICTIM WITH YOUR BARE HANDS until he is clear of the conductor, but DO NOT WASTE TIME.

2(a) Place the victim in the supine position.

2 (b) Keep the air passages clear by turning the head to one side, opening the patient's mouth and clearing it of water, saline, mucus or blood, a lot of which might have accumulated in the back of the throat.

(Figure 1)

3. 4. If the jaw is rigid try to force the mouth open by pressure on the gum behind the last molar tooth of the lower jaw. When the upper air passages are thus cleared, tilt the head backward and force the jaw forwards from the angles of the jaw in front of the ears. This would prevent mechanical obstructions to the upper air passages.

(Figure 2-3)

(a) Then hold the chin up and forward with one hand and pinch the nostrils of the victim with the other.

(Figure 4)

(b) Take a very deep breath and apply your mouth to that of the victim and blow into his mouth, until the chest of the victim moves up indicating filling of the lungs. (NEVER ALLOW THE CHIN TO SAG).

(Figure 4)

5. When the chest has moved up, withdraw your mouth and allow the chest to sink back. REPEAT this process every three to four seconds until the victim begins to breathe again or until he is taken over by a medical attendant. This method can be continued in an ambulance during transit of the patient from the site of accident to the nearest medical centre. HAVE SOMEONE ELSE AND SEND FOR A DOCTOR. KEEP PATIENT WARM AND LOOSEN HIS CLOTHING. DO NOT GIVE LIQUIDS UNTIL THE PATIENT IS CONSCIOUS.

IMPORTANT SAFETY INSTRUCTIONS

The equipment described in this manual is intended for use only by authorized personnel who are sufficiently trained in the installation and / or operation of electrical equipment.

Please read carefully all of the instructions listed and keep the manual for later reference.



- a) This equipment is designed to operate from a 220 Volt AC power supply. Never connect the equipment to any other voltages.
- b) Do not allow anything to rest or trap the power cord. Do not locate the equipment where persons could walk on the power cord.
- c) Remove the power to the equipment before cleaning. Do not use liquid cleaners or aerosol cleaners. Use a damp cloth for cleaning.
- d) Never open the equipment and change cards or modules with the power connected.
- e) Always use proper ESD protection when performing maintenance inside the equipment.

DEMOLITION OF MATERIAL TO PREVENT ENEMY USE

(A) General

The demolition procedure outlined below will be used to prevent the enemy from using or salvaging this equipment. **DEMOLITION OF THE EQUIPMENT WILL BE DONE ONLY ON ORDER OF THE COMMANDER.**

(B) Methods of Destruction

Use any or all the following methods to destroy the equipment.

- | | | |
|------------|---|--|
| Smash | - | Smash the crystals, controls semiconductors, coil, switches, transformers and handset. Use sledges, hand axes, pickaxes, hammers crowbars and heavy tools. |
| Cut | - | Cut the cords, headsets and wiring. Use hands axes or hatchets. |
| Burn | - | Burn the cords, resistors, capacitors, coils, wiring, technical manuals. Use gasoline, kerosene, oil, flamethrowers or incendiary. |
| Bend | - | Bend the panels, cabinet and chassis. |
| Explosives | - | If explosives are necessary, use firearms, grenades or TNT. |
| Disposal | - | Bury or scatter the destroyed parts in slit trenches, fox holes or throw them into streams. |

CHANGE NOTICE

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Personal Notes

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1 Introduction

1.1 General

This Operator Manual is intended for the users of the RFT1001 field telephone.

1.1.1 The Manual

The manual is divided into several chapters, each covering different aspects of the field telephone.

Chapter 2 : Contains brief technical data.

Chapter 3 : Overview of Field Telephone RFT1001

Chapter 4 : Describes the un-packing and removal of preservatives along with station list.

Chapter 5 : Describes the preparation for use & installation specifications of RFT1001.

Chapter 6 : Sequence of Operations.

Chapter 7 : Describes servicing and maintenance instructions.

Note: The product pictures, shown in various figures in this document, are just for representation purpose. Actual product may be little different than the product shown in figures.

1.2 Abbreviations

AC	Alternative Current
BAT MON	Battery Monitoring
CBP	Common Battery Pulse
CBT	Common Battery Tone
CNR	Combat Net Radio
DC	Direct Current
DTMF	Dual Tone Multiple Frequency
EMC	Electromagnetic compatible
EMI	Electromagnetic Interference
ESD	Electrostatic discharge
IDP	Inter Digit Pause
LB	Local battery (Magneto Mode)
LBR	Local battery with radio set
LED	Light Emitting Diode
MTBF	Mean time between failures
MTTR	Mean time to repair
PBX	Private Branch Exchange
PPS	Pulse per Second
PTT	Press to Talk
RH	Relative Humidity
RI	Ring Indication

2 Technical Data

2.1 General Information

The RFT1001 is a lightweight, ruggedized and portable field telephone. It can function as a normal magneto telephone (LB phone), capable of working with another magneto telephone of similar kind, or with a magneto exchange. It can also work as an auto telephone (CB phone) with dialling facility (either in PULSE or DTMF). The field telephone has a built-in dial keypad. In magneto phone mode, the field telephone works from 3 nos. of 1.5 Volt standard D type dry batteries.

2.2 Salient Features

- a) Compact, lightweight and ruggedized painted with olive green colour.
- b) Easy to install, operate and maintain.
- c) Operates from 3 nos. of 1.5V, 'D' type dry battery.
- d) Battery consumption under normal usage is minimal.
- e) Telephone is protected against reverse battery connections.
- f) The field telephone has a built-in dial keypad.
- g) The dial keypad is disabled during magneto mode (LB) and Radio mode (RAD).
- h) Handset of telephone uses Noise cancelling for improved speech communication in places with high ambient noises.
- i) The design of handset is user friendly and can be comfortably used while wearing helmet. The handset is held on telephone by use of spring-loaded clamps.
- j) For the easy maintenance purpose for a lineman, following visual indications are available in telephone set.
 - Visual indication when line is short.
 - Visual indication for battery health.
 - Visual indication for incoming ring.
 - Visual indication for outgoing ring.
 - Visual indication for reversed polarity of battery connection
- k) Can be used with similar field telephone or any magneto telephone and magneto exchange.
- l) Can be used on auto exchange lines.
- m) Dial keypad supports both PULSE and DTMF dialling.
- n) Dynamic receiver and microphone for high quality performance.
- o) Visual ringing indicator for incoming ring with auto exchange mode.
- p) Visual ringing indicator for incoming and outgoing ring with CBT, CBP & LB mode.
- q) Push button operation for magneto calling.

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- r) Ringer volume can be set to MUTE, LOW & HIGH by adjusting the SPEECH switch.
- s) Ringing tone is loud enough to be heard upto max 0f 3 Meters from telephone in ambient noise conditions.
- t) Line terminals and Battery terminals are spring loaded type for easy access but at the same time protected against damage due to accidental dropping of the telephone.
- u) Visual indication for battery status.
- v) Carrying pouch is suitably ruggedized as per military standards. It is olive green and has an adjustable strap for easy carrying. The telephone can also be operated without taking it out from the carrying pouch. The telephone can easily be removed from the case without the use of any special tools.
- w) The mechanical design of the telephone comprises the captive screws to avoid the loss of screws during field maintenance.



Figure 1: General View of Field Telephone, RFT1001



Figure 2: Carrying Bag for Field Telephone, RFT1001



Figure 3: Top View of Field Telephone, RFT1001

2.3 Mechanical Specifications

S/N	Specification	Range
1	Weight (In Kg)	a) 2.14 Kg with battery & handset b) Less than 1.8 Kg with handset and carry bag (w/o battery)
2	Dimension (in mm) Max	Height : 117 ± 1 mm Width : 124.50 ± 0.5 mm Length : 237 ± 1 mm

Table 1: Mechanical Specifications of RFT1001

2.4 Electrical Specifications

S/N	Specification	Range
1	Mode of Operation: a) CBT b) CBP c) LB d) RAD	a) Central Battery with DTMF dialling b) Central Battery with Pulse dialling c) Local Battery (Magneto mode) d) To connect with CNR Radio through external device RAU (optional)
2	Power Specs- a) Internal Battery b) Max Current Drain c) Idle current d) Max speech current e) Ext battery voltage	a) 4.5 Vdc, 3x1.5V 'D' type dry battery b) 180 mA (max) c) Nil d) 16 mA e) 4.5 Volt DC
3	Internal Battery Monitoring	Good, Poor & Reverse *Reverse is only for internal batteries
4	Loop Resistance a) In CBT/CBP Mode b) In LB Mode c) In RAD Mode (Loop Test)	a) 800 Ohms (max.) line loop (with 48V line voltage) b) 7500 Ohms (max.) line loop. c) 360 to 440 Ohms across line terminals when PTT is pressed.
5	Speech Range	55 dB
6	Speech Output (La – Lb terminals)	More than 900 mV.
7	Side Tone	300 mV (min) below Transmit level (Speech Output).
8	Receiver Frequency	0.3 Khz to 3.4 Khz
9	Ringing Voltage (LB Mode)	a) 90 V (minimum) at no Load b) More than 18V with $2K\Omega$ Load
10	Receive Efficiency	Not less than 350 mV
11	Send efficiency	Not less than 900 mV
	300 Hz	Should not less than 450 mV

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S/N	Specification		Range
	Freq Response	500 Hz	
		2000 Hz	
		3000 Hz	
	Side Tone Efficiency		At least 300 mV below the value of speech output test.
12	Signalling characteristics a) In CBT & CBP b) In LB		a) PPS : 10 ± 0.5 IDP : $800 \pm 10\%$ Flash : $300/600 \pm 20$ msec Make/Break : 60 % – 70 % b) Ring Freq : $25 \text{ Hz} \pm 2 \text{ Hz}$
13	Insulation resistance		Better than $5 \text{ M}\Omega$ at 250 V DC across body and line terminals
14	Lineman test facility		LED indication for following condition a) Line Open b) Line Short

Table 2: Electrical Specifications of RFT1001

2.5 Environmental Characteristic

1	Operating temperature	$-20^{\circ}\text{C} \pm 3^{\circ}\text{C}$, RH-95% to $+55^{\circ}\text{C}$ for duration 16 Hours
2	Storage temperature	$-30^{\circ}\text{C} \pm 3^{\circ}\text{C}$, RH-95% to $-70^{\circ}\text{C} \pm 2^{\circ}\text{C}$ for duration 16 Hours
3	Relative Humidity	0 – 95 % below $+30^{\circ}\text{C}$
4	High altitude	Up to 15000ft meter above sea level at $-20^{\circ}\text{C} \pm 3^{\circ}\text{C}$ for the duration of 16 Hours
5	Driving rain test	Static pressure: 200Kpa, for 1 Hour
6	Immersion	Depth 15 cm for 2 hours.

Table 3: Environmental Characteristics of RFT1001

3 Overview of Field Telephone RFT1001

3.1 About RFT1001

The RFT1001 is a lightweight, ruggedized and portable field telephone which is capable of use under all types of weather, terrain and other environmental conditions without any compromise on the performance of the equipment.



Figure 4: Isometric View of Field Telephone, RFT1001

The field telephone incorporates the latest techniques of transmission, signalling and other facilities. The field phone is capable to perform in any of the mode as given below;

- Auto telephone (Common Battery Mode in Tone - CBT)
- Auto telephone (Common Battery Mode in Pulse - CBP)
- Magneto telephone for point-to-point call or with another magneto exchange (Local Battery Mode-LB)
- Local Battery Radio Mode (RAD)

The field telephone works from 3 nos. of 1.5-volt standard 'D' type dry batteries.

3.2 Controls, Terminals and LEDs on RFT1001

Figure 5 and Figure 6 illustrate various controls, terminals, keys and buttons on the field telephone.

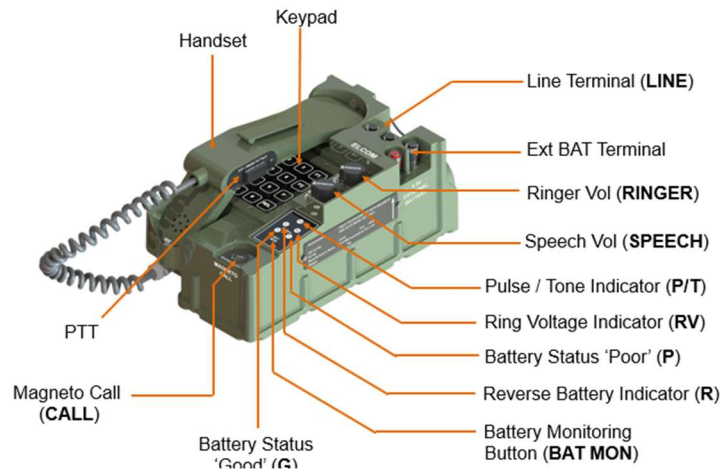


Figure 5: Controls, Terminals & Indicators

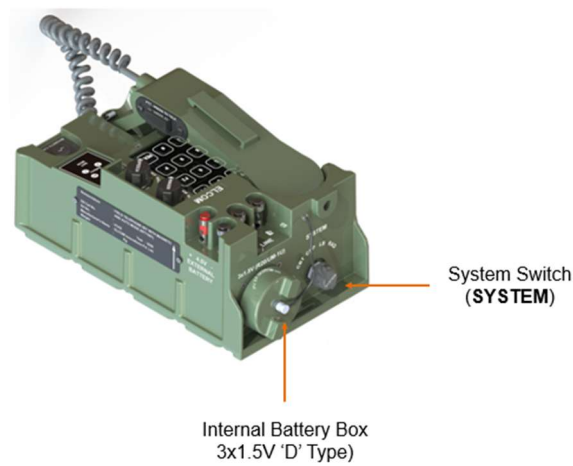


Figure 6: SYSTEM Switch & Battery Box of RFT1001

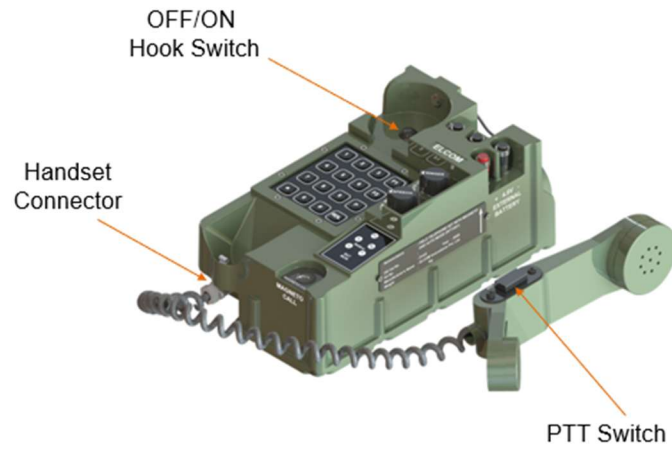


Figure 7: Handset Connector, Hook Switch & PTT

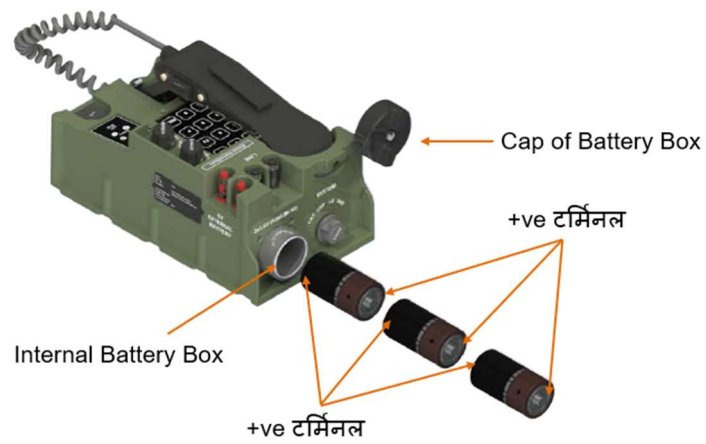


Figure 8: Internal Battery Box with 3x1.5V 'D' Type Batteries

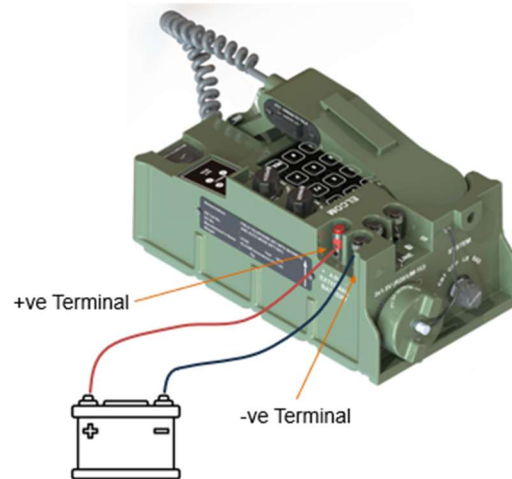


Figure 9: External Battery Connection

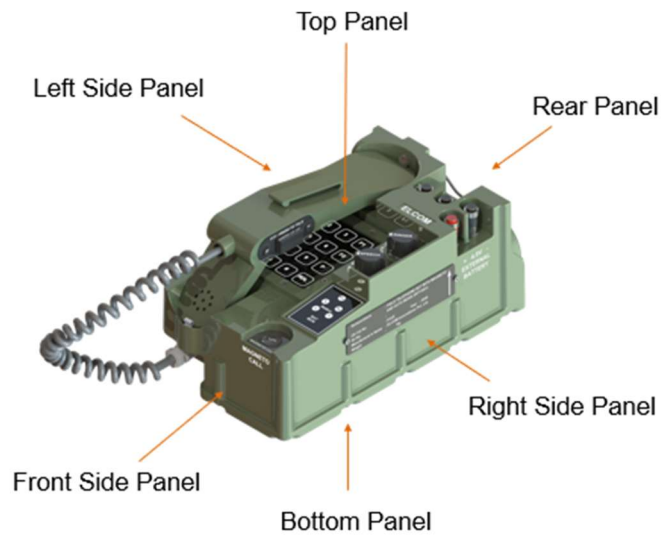


Figure 10: RFT1001 - Top, Bottom & Side Panells

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A ready reference table for the controls, terminals and indicators is given in Table 4. For the ease of understanding, all 6 panells are shown in Figure 10.

Controls, Terminals & Indicators	Function	Location	Mode
LINE Terminals	To terminate the line wires (from CB / LB exchange or LB telephone or Radio)	Top Panel	All modes
EXTERNAL BATTERY	To connect the external battery	Top Panel	LB, RAD
‘RINGER’ Switch	To set ringer sound output to MUTE, LOW or HIGH	Top Panel	CBT, CBP, LB
‘SPEECH’ Switch	To set Incoming voice level to Low, Medium & High.	Top Panel	All modes
Pulse/Tone (P/T) Indicator	To show whether RFT1001 is set to DTMF or Decadic	Top Panel	CBT & CBP
‘RV’ Ring Voltage Indicator	To indicate the incoming ring	Top Panel	CBT, CBP, LB,
‘BAT MON’ key	To monitor the Battery condition (Good / Poor)	Top Panel	-
‘G’ Good Battery Status	To indicate battery status ‘Good’ when Bat. Mon. switch is pressed	Top Panel	-
‘P’ Poor Battery Status	To indicate battery status ‘Poor’ when Bat. Mon. switch is pressed	Top Panel	All modes
‘R’ Reverse Polarity of Int Batteries	To indicate that the internal dry batteries are not inserted correctly in battery box.	Top Panel	All modes
‘MAGNETO CALL’ Button	To make a call to interfaced magneto phone or to magneto exchange.	Top Panel	LB Mode
Keypad	Keypad to dial the desired numbers	Top Panel	CBT, CBP
‘SYSTEM’ Switch	To select the mode of operation - CBT, CBP, LB or RAD	Rear Panel	-
Handset Connector	To connect the handset assembly	Front Side Panel	All modes

Controls, Terminals & Indicators	Function	Location	Mode
OFF/ON Hook Switch	It is used to detect the OFF-ON Hook action. It remains covered by the speaker of Handset.	Top Panel	CBT & CBP
PTT (Press to Talk) SWITCH	To activate the microphone circuit when pressed while talking.	On the Handset	All modes
Internal Battery Box	For inserting 3 nos. batteries (1.5 Volt 'D' Type) required for the telephone operation	Rear Panel	LB & RAD

Table 4: Control, Terminals & Indicators of RFT1001

3.3 Description of Controls, Terminals & LEDs

3.3.1 LINE - Line Terminals

In CBT, CBP, LB and RAD mode, the field wires are to be terminated in the line terminals. Line wires will have 2 wires which can be connected to Line Terminals in any of the polarity. Figure 5 shows the terminals for terminating the line.

3.3.2 External Battery Terminals

In case of non-availability of internal batteries, field telephone RFT1001 has a provision to connect an external source (4.5 Volt DC).

The connectivity of external batteries is polarity sensitive hence the +ve & -ve wires of external battery source (4.5Vdc) are to be connected to +ve & -ve terminals of EXTERNAL BATTERY as shown in the Figure 9.

3.3.3 RINGER - Ringer Volume Switch

The volume level of the buzzer sound can be adjusted by the RINGER switch. It is 3-position switch; MUTE, LOW & HIGH. In MUTE position, ringer will be Switched-Off.

3.3.4 SPEECH - Receive Volume Switch

RFT1001 has a incoming voice level control switch (SPEECH) to control the audio level of the handset. It is 3-position switch; LOW, MED & HIGH.

3.3.5 P/T - Pulse/Tone Indicator

This indicator shows whether the RFT1001 is set for DTMF dialling or Pulse dialling. The functionality of P/T indicator will be as explained in Table 5.

Position of SYSTEM Switch	Status of P/T Indicator	Description
CBT	ON	P/T indicator will remain ON all the time
CBP	OFF	P/T will remain ON & will flicker when a number is being dialled in Pulse mode.
LB	OFF	Not applicable.
RAD	OFF	Not applicable

Table 5: P/T Indicator

3.3.6 RV - Ring Voltage Indicator

The Incoming Ring Voltage indicator (RV) on the field telephone, shown in Figure 5, will indicate the presence of ringer voltage in the telephone line. This LED will light up when there is an incoming call to the telephone either in CBT or CBP or LB mode.

The indicator (RV) will also glow when a ringer voltage is sent down the line for making an outgoing call in the Magneto mode (LB) and LED will glow while press the PTT for transmission.

3.3.7 BAT MON - Battery Monitoring Buttnony

RFT1001 has a **BAT MON** key as shown in Figure 5, which is used to view the status of the internal battery. When BAT MON key is pressed, indicators G & P are used to know the battery status (refer chapter 3.3.8).

3.3.8 'G' & 'P' - Battery Status Indicators

On pressing **BAT MON** key, indicators 'G' & 'P' will show the battery status. The significance of these indicators is given in Table 6.

Status of G & P	Significance	Action to be taken
For Internal Batteries		
If 'G' LED is On & 'P' LED is Off.	Consolidated battery voltage is sufficient enough (i.e. > 3.5V)	No action to be taken
If 'G' LED is Off & 'P' LED is On	Consolidated battery voltage is not sufficient (i.e. < 3.5V)	Batteries to be replaced
For External Power Source		
If 'G' LED is On & 'P' LED is Off.	External Voltage is sufficient enough (i.e. > 3.5V)	No action to be taken
If 'G' LED is Off & 'P' is On	Connected power source is not sufficient (i.e. < 3.5V)	Either replace the power source or increase the voltage.

Table 6: Battery Monitoring

3.3.9 'R' - Reverse Battery Indicator

This indicator will be switched-on if internal batteries are inserted in reverse condition in internal battery box. To have proper functionality of RFT1001, it is essential to insert all 3 No. internal batteries in correct polarity as shown in Figure 8

3.3.10 MAGNETO CALL Button

When the field telephone is in LB mode, for making an outgoing call on the line, the **MAGNETO CALL** button has to be pressed for generating a ringing voltage and sending it across the line. The voltage will be generated as long as the **MAGNETO CALL** button is being pressed. Magneto Call Button is shown in the Figure 5.

3.3.11 Dial Keypad

The field phone has a dial keypad for dialling in DTMF and Pulse (in CBT & CBP mode), which can be used for dialling out any digits on the phone line. The keypad is a 4x4 matrix, which has several buttons as explained below;

- 0 to 9 - Dialling digits 0 to 9
- * & # - To dial * & # (*works only with CBT mode*)
- F1 - Flash key with flash time 300ms
- F2 - Flash key with flash time 640ms
- P - Pause Key with 2.2ms
- RDL - Redial Key to redial a last number dialled.

3.3.12 SYSTEM Switch

This is a 4-position rotary switch in the phone set as shown in Figure 6. The various operation modes can be set via this switch. The position is marked as shown below:

Positions marked as	Mode of operation
CBT	To set the phone to work in Central Battery mode with DTMF dialling.
CBP	To set the phone to work in Central Battery mode with Pulse dialling.
LB	To set the phone in Local Battery mode
RAD	To interface the phone with Combet Net Radio through external Radio Adoptor Unit

Table 7: SYSTEM Switch

3.3.13 Handset Connector

It is used to connect the handset to the RFT1001 as shown in Figure 7.

3.3.14 OFF/ON Hook Switch

OFF-ON Hook switch is shown in Figure 7. Through this switch only, RFT1001 detects whether handset is kept on phone or not. If handset is kept on phone properly, this position is called 'ON-Hook' & if handset is

not kept on phone, this position is called 'OFF-Hook'. Hence this switch is remained covered by the speaker of Handset.

3.3.15 PTT Switch

The Press-to-talk (PTT) switch is present on the handset of the field telephone. This switch is to be pressed to activate the microphone of the handset. This is applicable only in LB & RAD mode. There is no need to press PTT in CBT & CBP mode for making a conversation. PTT with handset is shown in the Figure 5 & Figure 7.

3.3.16 Internal Battery Box

The field telephone requires a bank of internal batteries (3x1.5V, 'D' type) for operating in LB and RAD mode only. This battery bank is used for generating the outgoing ringing voltage in the LB mode.

For internal battery bank, 3x1.5V 'D' type batteries are to be inserted into the battery box of the field telephone as shown in the Figure 8.

4 Unpacking & Removal of Preservations

4.1 How to Unpack

As the telephone set is of rugged nature, there will not be any need for special packing or preservatives. The unpacking of the equipment will therefore be of conventional nature.

Each telephone is packed in a carrying pouch which is made of anti-fungus material & it is finally packed inside a carton box.

It is highly recommended to keep all the packing material preserved so that whenever equipment is supposed to be packed again for storage or transportation, same packing material can be reused.

4.2 Station list

S. No.	Description	Manufacturer's Part No.	DS Cat No.	No. Off
(1)	(2)	(3)	(4)	(5)
1	Field Telephone RFT1001 with Handset	DPAR002014	-	1
2	Carrying Bag	P73000379	-	1
3	User Handbook	P72163028	-	1

Table 8: Station List

5 Installation

5.1 Preliminary Check

- a) The telephone has been designed to perform as ground equipment, exposed or unpacked.
- b) Place the telephone at a convenient place.
- c) Insert three numbers standard 'D' type dry batteries in the internal battery box.
- d) Check the battery status by pressing the **BAT MON** Key, refer chapter 3.3.7.
- e) If indicator '**G**' glows, battery voltage is Ok. If indicator '**P**' glows, replace all 3 internal batteries with new set of batteries.
- f) Press the **PTT** switch and blow into the microphone and listen for the side tone.

The telephone set is now ready to be installed in any of the mode explained below.

5.2 Auto Mode (CBT / CBP)

Figure 11 shows the connectivity diagram to install the field telephone in **CBT** or **CBP** mode.

In auto mode (CBT/CBP), the telephone is powered by common battery exchange and does not use its internal batteries.

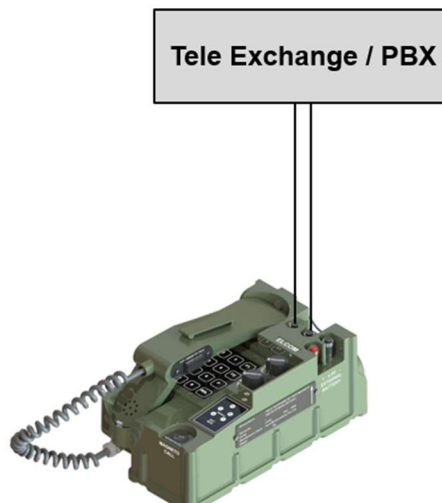


Figure 11: RFT1001 in CBT/CBP Mode

The following steps are to be followed for installation of the field telephone in Auto mode:

- a) Terminate the telephone line wires at the **LINE** terminals on the field telephone. Care should be taken while handling the telephone line wires as there is a possibility of presence of ringer voltage in the line being connected to the field telephone.
- b) For DTMF dialling, set the **SYSTEM** switch in **CBT** mode & for pulse dialling, set the **SYSTEM** switch in **CBP** mode.
- c) On lifting the handset from the cradle, dial tone coming from Tele Exchange / PBX will be heard in the receiver. The built-in dial keypad can now be used to dial the desired number.

The field telephone has been installed successfully.

5.3 Magneto Mode (LB)

As described earlier, the field telephone can function as a normal magneto telephone with a telephone of similar kind, or with a magneto exchange.

Figure 12 and Figure 13 shows the connectivity diagram to install the field telephone in LB mode with another magneto telephone and another magneto exchange respectively.

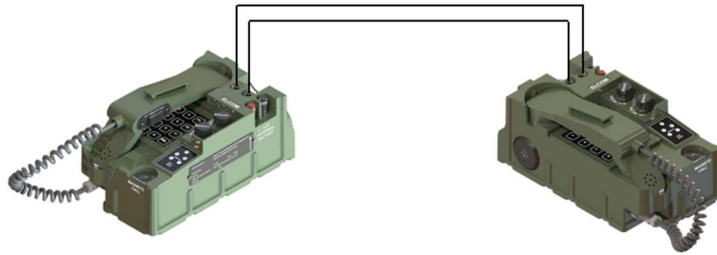


Figure 12: RFT1001 in LB Mode with another LB Phone

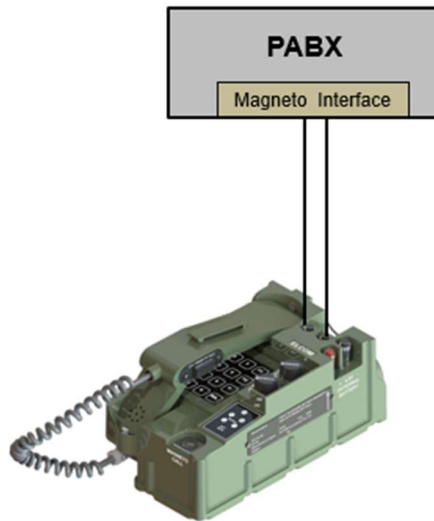


Figure 13: RFT1001 in LB Mode with Magneto Exchange

The following steps are to be followed for the installation of the field telephone in LB mode:

- a) Ensure that 3 numbers of standard 1.5 volt 'D' type dry batteries in the battery box are present.
- b) Check the battery status by following the procedure explained in the chapter 3.3.7.
- c) If external battery is to be used, connect the external battery source (4.5Volt DC) to the EXTERNAL BATTERY on the field telephone.
- d) The connectivity of external batteries is polarity sensitive hence +ve & -ve wires of external battery source (4.5V) have to be connected to +ve & -ve terminal of EXTERNAL BATTERY terminals respectively.
- e) Set the **SYSTEM** Switch to position **LB** for setting the field telephone in LB mode.
- f) Lift the handset, press the **PTT** switch and blow into the microphone and listen for the side tone. If the side tone is heard, the field telephone is now ready for use.
- g) Terminate the line wires coming either from the similar kind of magneto phone or from the magneto exchange at the LINE terminals on the field telephone. Line wires are polarity independent hence can be connected to the line terminals in any polarity.
- h) Care should be taken while handling the telephone line wires as there is a possibility of presence of ringer voltage in the line being connected to the field telephone.
- i) The field telephone is ready to use in magneto mode (LB).

5.4 RAD Mode - Magneto Mode with CNR Radio

The field telephone RFT1001 shall be used in **RAD** mode if the phone has to be interfaced with a CNR radio set through an external Radio Interface Module. Figure 14 shows the connectivity diagram to install the field telephone with CNR Radio. Following steps are to be followed for installation of the field telephone RFT1001 with CNR radio set:

- a) Insert 3 numbers of standard 1.5 volt 'D' type batteries in the battery box.
- b) Check the battery status by following the procedure explained in the chapter 3.3.7.
- c) If external battery is to be used, connect the external battery to the **EXTERNAL BATTERY** on the field telephone. Remember external battery source should be 4.5 Volt DC.
- d) Set the **SYSTEM** switch to position **RAD** for setting the field telephone in **RAD** mode with CNR radio set.
- e) Lift the handset, press the **PTT** switch and blow into the microphone and listen for the side tone. If the side tone is heard, the field telephone is now ready for use.

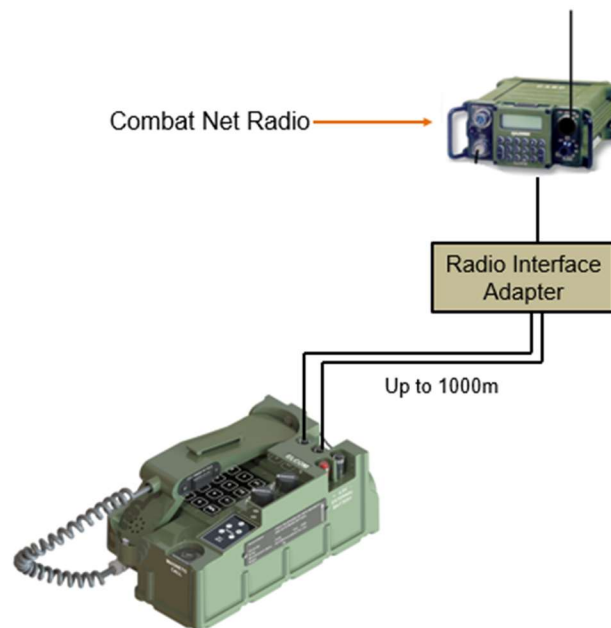


Figure 14: RFT1001 in RAD Mode

- f) Terminate the line wires coming from radio interface module at the **LINE** terminals on the field telephone. Line wires are polarity independent hence can be connected to the line terminals in any polarity.

The field telephone is ready to use in RAD mode with CNR radio set via radio interface module.

6 Operations

6.1 Auto Mode (CBT or CBP)

6.1.1 Outgoing Call

- a) Lift handset.
- b) Hear dial tone.
- c) Start dialling the desired number on the keypad.
- d) If the dialling is in pulse mode, indicator P/T will blink according to the number being dialled.
- e) If the dialling is in tone (DTMF) mode, indicator P/T will glow constantly.
- f) Wait for reply from the far end.
- g) On getting a reply, user can start the conversation.
- h) Once the conversation is over, put the handset on-hook.

6.1.2 Incoming Call

- a) Incoming call is indicated by a ringing tone in the field telephone.
- b) Ring Indicator (RV) will start glowing for visual intimation of incoming call in the telephone line.
- c) Lift handset to answer the incoming call and start speaking.
- d) Once the conversation is over, put the handset on-hook.

6.1.3 Hold a Call

- a) While talking on an outgoing or incoming call, if user wants to hold the call, user has to press F1 or F2 key. F1 key is to flash the line for 300ms & F2 is for 640ms.
- b) To unhold a held call, press RDL key again.
- c) Flash time has to be decided as per the auto exchange which is used to interface RFT1001.

6.1.4 Last Number Redial

- a) To dial the last number again, first release the RFT1001 by putting handset on hook.
- b) Lift the handset again & press RDL key.
- c) RFT1001 will dial the last number on its own.
- d) Wait for reply from the far end.
- e) On getting a reply, user can start the conversation.
- f) Once the conversation is over, put the handset on-hook.

6.2 Magneto Mode (LB Mode)

6.2.1 Outgoing Call

- a) Lift handset.
- b) Press the **CALL** button to send the calling signal.
- c) Ring indicator (RV) will start glowing, confirming that the ringing current is being sent down the telephone line.
- d) Wait for reply from the far end.
- e) On hearing a voice response, use PTT switch to start conversation.
- f) To listen the receiving voice, release the PTT button.
- g) Do not press the **CALL** button during conversation.
- h) Once the conversation is over, put the handset On-Hook.

6.2.2 Incoming Call

- a) Incoming call is indicated by a ringing tone in the field telephone.
- b) Ring Indicator (RV) will start glowing for visual intimation of incoming call in the telephone line.
- c) Lift handset to answer the incoming call.
- d) Press PTT to start conversation.
- e) To listen the incoming voice, release the PTT button.
- f) Do not press the **CALL** button during conversation.
- g) Once the conversation is over, put the handset On-Hook.

6.3 RAD Mode with Radio Set

6.3.1 Outgoing Call

- a) Lift the handset of the RFT1001 phone and press the PTT switch and start speaking.
- b) User's voice will be transmitted to the far end side.
- c) After speaking, release PTT switch and wait for the far end party to speak
- d) If the party speaks, listen the incoming voice.
- e) Once the conversation is over, put the handset On-Hook.

6.3.2 Incoming Call

- a) In RAD mode, there will not be any incoming ring. The telephone instrument remains in listen mode always and hence if there is any incoming voice, user will be able to listen in the handset. Remember PTT switch should be released when listening.
- b) To speak out, press the PTT button and start speaking. The user's voice will be transmitted to far end side.
- c) Once the conversation is over, put the handset On-Hook.

7 Servicing & Maintenance Instructions

7.1 Daily Maintenance

- a) Clean the field telephone with a piece of clean dry cloth to remove dust.
- b) Clean the area between the **LINE** terminals with a hard nylon brush and check the line wires for proper gripping with the line terminals.
- c) Clean the area between the **EXTERNAL BATTERY** terminals with a hard nylon brush.
- d) Disconnect the handset from Handset Connector & check the connector pins for any carbon deposits. If observed, clear it.
- e) Check the battery voltage by pressing **BAT MON** Key. If battery monitoring indicator 'P' lights up, replace all 3 batteries.
- f) Check the **PTT** by pressing it. A distinct click sound will be heard. Press **PTT** and blow into the microphone. Side tone will be heard in the receiver.
- g) Check the **MAGNETO CALL** button by pressing it. A distinct click sound will be heard.
- h) Connect a Magneto line. Press **MAGNETO CALL** button to call and talk with the other end's telephone by pressing PTT.
- i) Ask the other end's magneto telephone user to send a call signal and check if incoming call is detected by the field telephone.
- j) Connect an auto line. Lift the handset, hear the dial tone, dial a subscriber's number and talk.

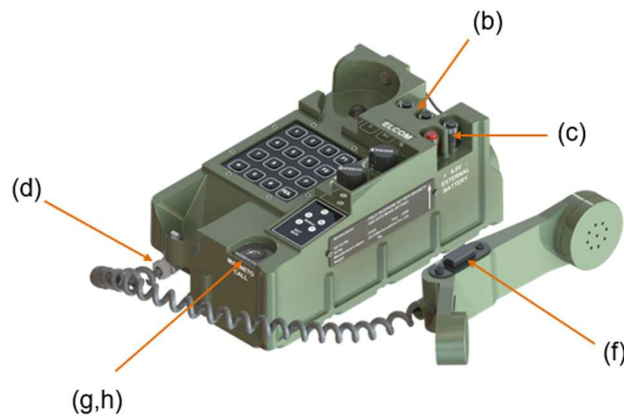


Figure 15: Daily Maintenance

7.2 Weekly Maintenance

- a) Repeat all the steps mentioned in the Daily maintenance routine. In addition, do the following:
- b) Open the battery box cap & takeout all 3 dry batteries from the battery box.
- c) Clean the metal contacts of all 3 batteries with a hard nylon brush.
- d) After cleaning, put the batteries back in the battery box & tight the cap properly.
- e) While putting the batteries inside the internal battery box, take extra precautions for the polarities of the batteries. Refer Figure 8.

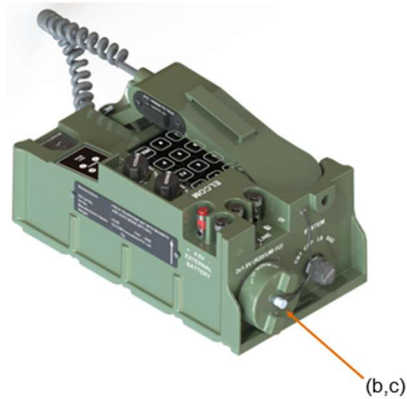


Figure 16: Weekly Maintenance

7.3 Monthly Maintenance

- a) Repeat all the steps mentioned in the daily maintenance routine and the weekly maintenance routing. In addition, do the following:
- b) Connect an AC voltmeter across the line terminals as shown in the Figure 17.
- c) Set the **SYSTEM** Switch in LB mode and press **MAGNETO CALL** button.
- d) AC voltage measured should be more than 90 volts.

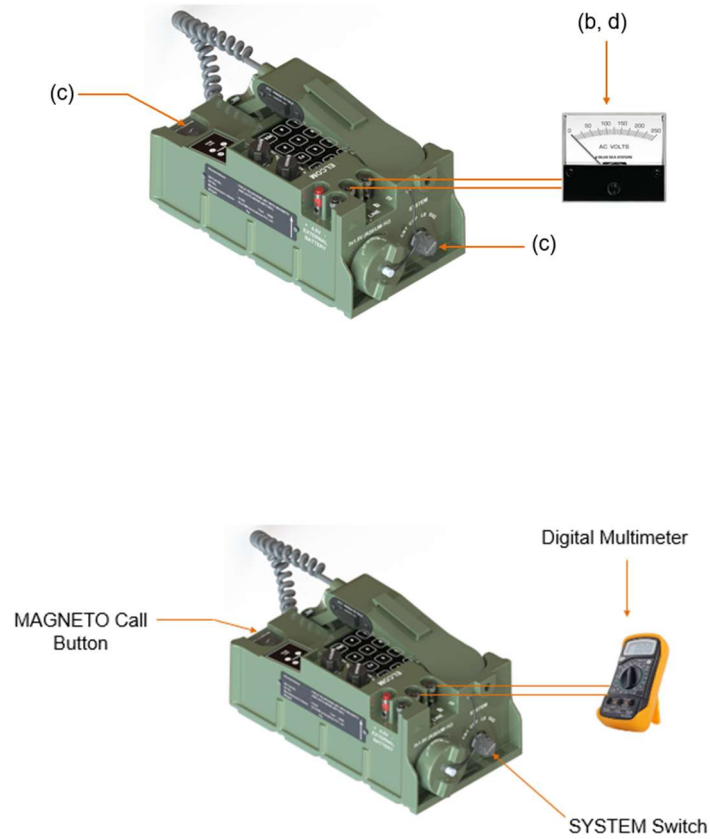


Figure 17: Monthly Maintenance

Note: If the RFT1001 is not in use for more than one month, remove all 3 batteries from the internal battery box to avoid damage caused by battery leakage.

7.4 General Fault Finding

Fault	Response	Action
No Incoming Ring	Buzzer will not ring	a) Check for the volume switch; move the knob to MED or HIGH level. b) Check for Ringer Indicator; if it is glowing, check the telephone Buzzer.
LB / RAD mode not working properly	Not able to make outgoing call in LB / RAD mode	a) Verify the Battery status by pressing the BAT MON button. b) Change the same if found faulty.
Line wire short	No incoming or outgoing in any of the mode.	a) If the line wires are shorted in the field, R visual indication LED will permanently glow b) Try to remove the line shortage and verify the communication
Noise during conversation	Noisy communication	Check coil cord of the handset.
No speech from handset	-	a) Check the coil cord connectivity. b) Put the phone in LB mode and press the PTT, blow on handset microphone, if some part of the blow is there, handset is OK, else change the same.

Table 9: General Fault Finding

Note: If after doing above debugging, RFT1001 is not functioning as expected, RFT1001 need to be sent to base workshop for repair.

8 Document Change History

Rev.	Change	Date	Author	Approved by
PA1	Initial Document	5/10/24	SKS	PS
PA4	First release	15/02/26	SG	AS
PA8	Figures updated for HS connector	21/04/26	SG	AS

Table 10: Document Change History

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